



CENG 408

Innovative System Design and Development II
2025-2026 Spring

**MULTI-LABEL CHEST X-RAY DISEASE
CLASSIFICATION AND EXPLAINABILITY
WITH DEEP LEARNING**

CONCLUSION & FUTURE WORK

YAREN AKDOĞAN 202111046

SEZİN KÖROĞLU 202211046

İBRAHİM TAŞPINAR 202111072

CEVDET ALP ÖZGÜN 202211406

İPEK BAŞ 202211013

SALİH BARKIN AKKAYA 202211004

Table of Contents

1.	Model Architecture and Implementation	3
1.1.	DenseNet-121 Backbone	3
1.2.	Convolutional Block Attention Module	3
1.3.	Lessons Learned.....	3
1.1.	Future Work	4
2.	Visualization with GradCAM++	4
2.1.	Attribution Mechanics and Hook Implementation.....	4
2.2.	API Segmentation and Contour Generation	5
2.3.	Lessons Learned.....	5
2.4.	Future Work	6
3.	Model Training & Dataset.....	6
3.1.	Lessons Learned.....	6
3.2.	Future Work	7
4.	Frontend.....	8
4.1.	<i>Role Selection</i>	8
4.2.	Doctor Registration and Pending Approval Mechanism.....	9
4.3.	Doctor Registration and Pending Approval Mechanism.....	10
4.4.	End-to-End Clinical Workflow and Patient Management.....	11
4.5.	Real -Time Neural Processing and Feedback Mechanisms	12
4.6.	Explainable AI (XAI) and Diagnostic Visualization	12
4.7.	Seamless Diagnostic Reporting and Patient Record Management	13
4.8.	Integrated Digital Archiving and Diagnostic History	14
4.9.	Lessons Learned.....	15
4.10.	Future Work	15
5.	Database Design and Implementation	16
5.1.	Applications.....	17
5.2.	Lessons Learned.....	18
5.3.	Future Work	19

1. Model Architecture and Implementation

The final model was comprised of DenseNet-121 along with Convolutional Block Attention Module (CBAM). This combination helped the model achieve high performance across different thoracic diseases.

1.1. DenseNet-121 Backbone

We have chosen DenseNet as the backbone of our model due to its proven success in medical image classification tasks. The dense connectivity technique used in DenseNet allows the outputs of previous layers to be utilized in all future layers within a dense block in the model, which mitigates the possible vanishing gradient issues we may encounter. The segmented structure of DenseNet allowed us to insert the necessary models without disturbing the stability of our model successfully.

1.2. Convolutional Block Attention Module

We successfully placed CBAM modules directly after each of the four dense blocks. Because CBAM does not change the shape or size of the layers, it fit into the existing network without requiring any major structural modifications.

- **Channel Attention:** This part helped the model decide what features mattered most. By using both average and max pooling and passing them through a shared neural network layer, the model learned to prioritize the most important details for identifying each disease.
- **Spatial Attention:** This part helped the model locate where those important features were on the X-ray. It highlighted specific areas and ignored irrelevant parts of the image.

1.3. Lessons Learned

- **Feature Sharing and Flow:** Using DenseNet showed that passing information from early layers directly to later ones is very effective for medical images. This design helped the model keep track of details that would usually be lost in a normal CNN.
- **System Modularity:** The segmented structure of the backbone allowed us to treat the model as distinct pieces, making it simple to add the CBAM modules between blocks without needing to change the core layers or the final output setup.
- **Combining Attention Types:** We found that using both channel and spatial attention together provided the best results. While the backbone handles general feature extraction, the attention modules act as a filter to help the model focus on important textures while ignoring irrelevant parts of the X-ray.

1.1. Future Work

For the further stages of this research, we plan to improve the model's reliability and performance. A primary focus will be to refine the internal attention mechanisms. We intend to conduct ablation studies to measure the specific impact of the Channel and Spatial attention components on our final results. This will help us determine which module contributes most to the model's accuracy and allow us to optimize the architecture by focusing on the most effective features.

We also aim to improve the model's efficiency. By experimenting with the growth rate of the DenseNet blocks, we hope to create a more lightweight version of the architecture. This change would allow the model to run on lower-power devices or tablets without a significant loss in performance.

Additionally, we intend to explore architectural changes that account for the relationships between different output categories. By helping the network understand which features typically appear together, we can improve the logical consistency of the model's predictions. Finally, we plan to better align the internal spatial attention maps with the explainability layers to ensure the model's focus consistently matches the targeted regions.

2. Visualization with GradCAM++

To provide explainability to our model outputs, we used GradCAM++. This method identifies the specific regions within the chest X ray that influence the classification decisions of the model. Unlike standard GradCAM, the GradCAM++ variant utilizes a higher order derivative weighting formulation that handles multiple occurrences of a pathology in a single view with superior localization meaning we can display multiple effected regions at once.

2.1. Attribution Mechanics and Hook Implementation

The localization process maps the internal representations learned by the network directly to the spatial layout of the input image. To implement this without modifying the forward inference, execution hooks are dynamically attached to the second last convolutional layer of the backbone model. During execution, the forward hook captures the feature activation maps as they flow through the network.

When the model predicts a specific thoracic disease, the system runs a backward pass to see how changes in the final prediction score relate to the captured features. It calculates a weighting factor “alpha” using the second and third derivatives of this score. This keeps the system accurate even if the same disease shows up in separate parts of the lung at the same time. Finally, the system multiplies these weights by the feature maps and passes them through a ReLU function. This filters out negative data, leaving only the visual features that directly confirm the presence of the disease.

2.2. API Segmentation and Contour Generation

To connect the raw heatmaps with the user interface, the backend application runs an endpoint that processes pixel data into clear shapes. When a disease probability passes its diagnostic threshold, the system converts the blurry attribution map into a sharp binary mask.

This mask is then scaled to a standard resolution using bilinear interpolation. Next, a marching squares algorithm traces the outer edges of the area, creating continuous coordinates that form completely hollow outlines. Because these coordinates map out only the thin perimeter instead of a filled, solid shape, the final user interface can draw the borders without hiding the underlying lung tissue and bones. Finally, these hollow contour lines are turned into JSON lists and sent to the client application, allowing the interface to show clean borders.



Figure 1: Original Xray with Cardeomelagy and the GradCAM++ Output

2.3. Lessons Learned

- **Mathematical Stability in Backpropagation:** Utilizing higher order gradients through GradCAM++ provides stable localization metrics for multi label tasks. Standard gradient weighting often fails when multiple independent pathologies overlap in the same spatial plane.
- **Vector Contours over Heatmaps:** Transforming raw pixel arrays into hollow vector contours significantly reduces the data payload transmitted over network protocols. It allows client endpoints to overlay diagnostic visuals without rendering performance bottlenecks or blocking key medical details.
- **Threshold Dependency for Extraction:** Applying fixed thresholding to mask generation can introduce structural noise if the model baseline shifts. Vector boundaries are highly dependent on the accuracy of the preceding activation amplitudes.

2.4. Future Work

For the further stages of this research, we plan to enhance the visualization pipeline to maximize its clinical clarity. A primary focus will be replacing the current binary mask thresholding method with an adaptive contour extraction algorithm that computes localized gradient variations. This would ensure that boundaries remain structurally accurate even for diffuse pathologies like Infiltration or Edema.

We also aim to reduce the computational overhead of the backpropagation sequence. By implementing an optimized tracking system for the hooks, we intend to compute multiple class attribution maps during a single backward pass. Finally, we plan to validate the structural alignment between these generated visual boundaries and the internal attention maps of the network to confirm that the explanation layer truly reflects the decision paths of the model.

3. Model Training & Dataset

The NIH ChestX-ray14 dataset used in this study contains 112,120 frontal chest X-ray images labeled for 14 different thoracic diseases. Although the original dataset includes both posterior-anterior (PA) and anterior-posterior (AP) views, this project specifically filtered the data to focus only on PA views to optimize hardware constraints and training efficiency. To manage the significant class imbalance within the dataset weighted learning strategies were prioritized. Several image processing scenarios were initially designed, including intensity enhancement, bone suppression, and Butterworth band-pass filtering, to improve contrast and structural visibility. These steps prepared the standardized 224x224 RGB images required for the deep learning architecture.

This project developed a deep learning system for multi-label chest X-ray classification using the NIH ChestX-ray14 dataset. The development process followed an evolutionary path across four models. Initially, a baseline model using BCEWithLogitsLoss was created, which showed high overall accuracy but failed to detect rare diseases due to class imbalance. To solve this, Asymmetric Loss (ASL) was implemented, which significantly improved the detection of rare pathologies like Hernia and Pneumothorax. While experiments with image preprocessing (CLAHE and Gamma Correction) showed that artificial enhancements can introduce noise and reduce performance for the CBAM architecture, the final model successfully maximized diagnostic power by using Dynamic Class-Specific Thresholds. This final iteration achieved a Macro F1 score of nearly 0.36, approaching the level of human inter-radiologist agreement for this dataset.

3.1. Lessons Learned

- **Threshold Flexibility:** Fixed decision thresholds (0.50) are not effective for medical diagnosis. Each disease requires a specific threshold based on its visual characteristics and prevalence.

- **Data Integrity:** Aggressive image processing can destroy subtle natural features that the model needs for accurate classification.
- **Metric Selection:** In imbalanced medical datasets, Macro F1 and AUC are more reliable indicators of performance than simple accuracy.
- **Input Resolution Constraints:** Reducing images to 224x224 pixels makes it difficult for the model to detect fine pathological details. Small features like nodules can be lost during downscaling, indicating that higher resolutions are better for diagnostic tasks.
- **Attention Mechanism Sensitivity:** The CBAM architecture is highly sensitive to changes in pixel intensity. Artificial enhancements like CLAHE can cause the attention layers to focus on noise or artifacts instead of real anatomical structures.

3.2. Future Work

For the further stages of this research, we plan to increase the technical goals to improve the accuracy and clinical use of the system. A primary focus will be to improve the dataset. The current model was trained only on PA (Posterior-Anterior) views because of hardware limits and training time. In the future, we aim to include AP (Anterior-Posterior) views to make the model more reliable. Because these two positions change how the body appears on the X-ray, training the model to recognize both views will improve the diagnosis.

We also plan to test different loss functions. While Asymmetric Loss (ASL) managed the class imbalance in the final model, it does not consider the connections between different diseases. We intend to explore loss functions like Focal Loss or Label Correlation Loss. These functions help the model understand which diseases usually happen together, which should reduce false positives and make the predictions more logical.

Finally, we will update the data preparation. We already use augmentations like 7-degree rotations and horizontal flips to prevent overfitting. In future iterations, we will move the image processing methods used in this project, such as CLAHE and Gamma correction, from static preprocessing to dynamic augmentation during training. This approach will help the model handle different image qualities and noise without losing important medical signs. This change aims to make the model more stable across different hospital hardware and real-world settings.

4. Frontend

4.1. Role Selection

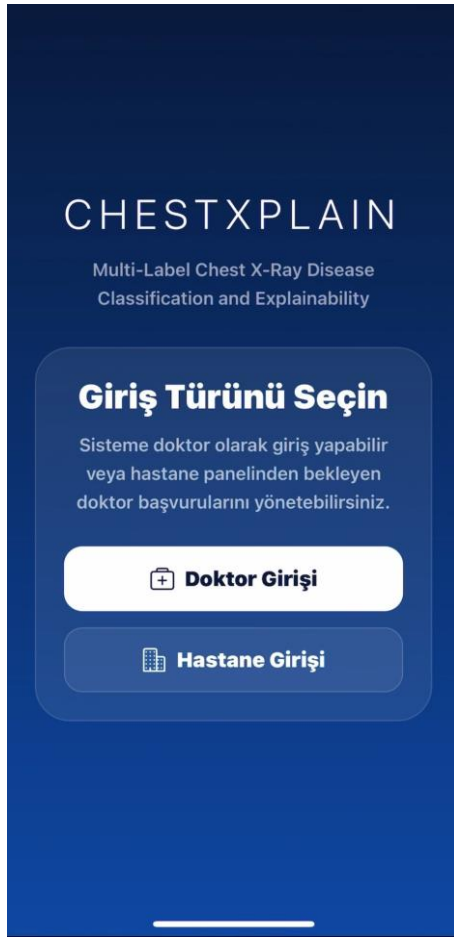


Figure 2: Initial Role Selection Screen

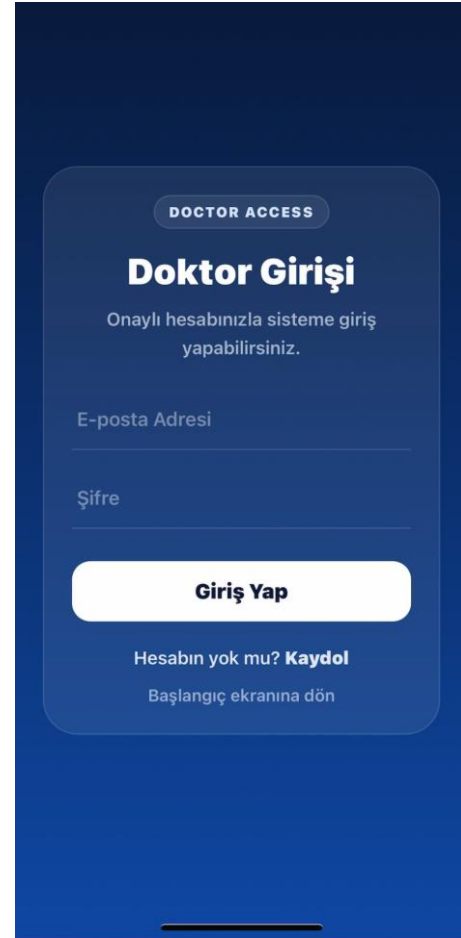


Figure 3: Doctor Login Interface

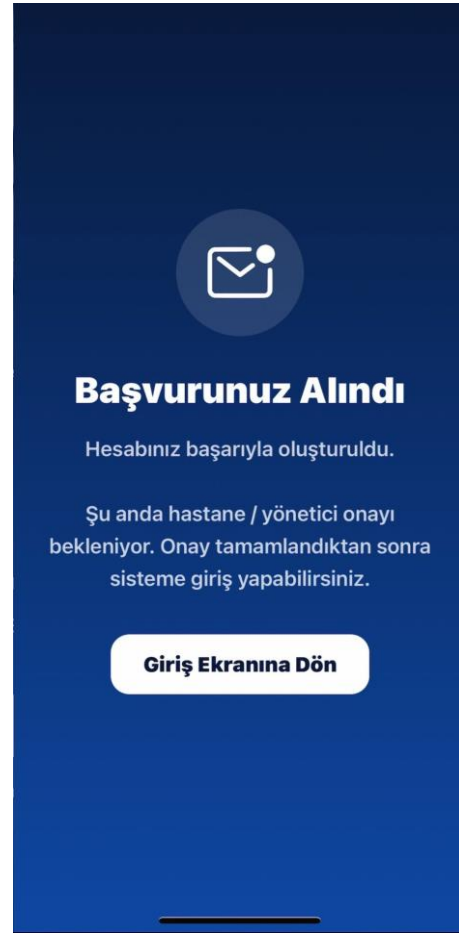
The frontend workflow begins with a role-based entry screen that clearly separates Doctor Access from Hospital Access. This initial decision point is essential for preserving the institutional logic of the system, because doctors and hospital administrators are responsible for different operations. As shown in Figure 2, the mobile interface introduces the application and directs users to the correct authentication path. Figure 3 presents the doctor login screen, where previously approved physicians can securely enter the system using their e-mail address and password. This separation of roles at the interface level improves usability, reduces ambiguity, and ensures that each user is guided toward the correct workflow from the very first interaction.

4.2. Doctor Registration and Pending Approval Mechanism



The screen features a background image of a doctor in a white coat with a stethoscope. The title "DOKTOR KAYIT" is prominently displayed. Below it, a subtitle states: "Başvurunuz hastane onayından sonra aktif olacaktır." The registration form includes fields for "Ad Soyad", "E-posta Adresi", "Şifre", "Görev Yaptığınız Hastane", and "Hastane Seçiniz...". A "Hesabı Oluştur" button is at the bottom, followed by a link "Zaten hesabın var mı? Giriş Yap".

Figure 4: Doctor Registration Screen



The screen has a solid dark blue background. At the top, there is an icon of an envelope with a checkmark. The title "Başvurunuz Alındı" is centered. Below it, the text reads: "Hesabınız başarıyla oluşturuldu." and "Şu anda hastane / yönetici onayı bekleniyor. Onay tamamlandıktan sonra sisteme giriş yapabilirsiniz." A "Giriş Ekranına Dön" button is located at the bottom.

Figure 5: Pending Approval Confirmation Screen

To support controlled onboarding, the application includes a dedicated doctor registration module. As illustrated in Figure 4, a new physician is required to provide personal credentials together with the hospital to which they belong. This design decision is especially important because the system does not treat doctors as independent users; instead, each account is institutionally linked to a hospital profile. After the registration request is submitted, the frontend immediately informs the user that the account has been created but is not yet active. As shown in Figure 5, the physician is explicitly notified that hospital or administrator approval is required before login is allowed. This intermediate confirmation screen is a critical part of the clinical workflow because it establishes a controlled approval mechanism rather than allowing unrestricted direct access.

4.3. Doctor Registration and Pending Approval Mechanism

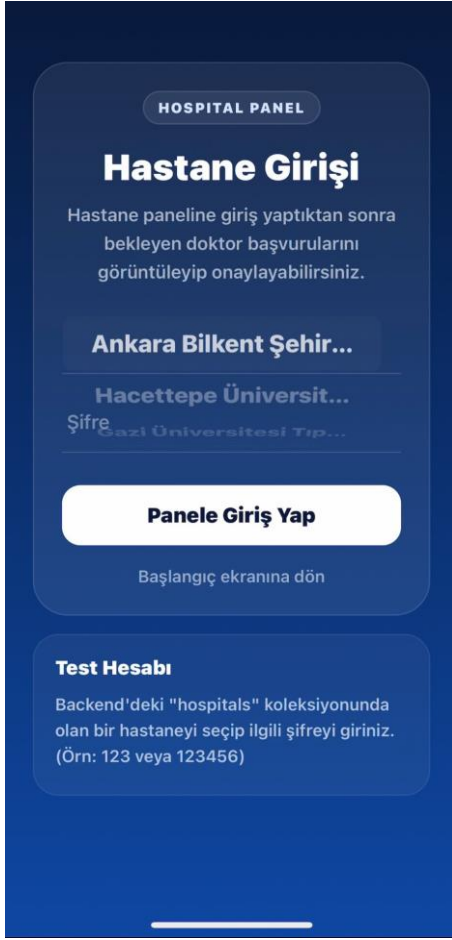


Figure 6: Hospital Login Interface

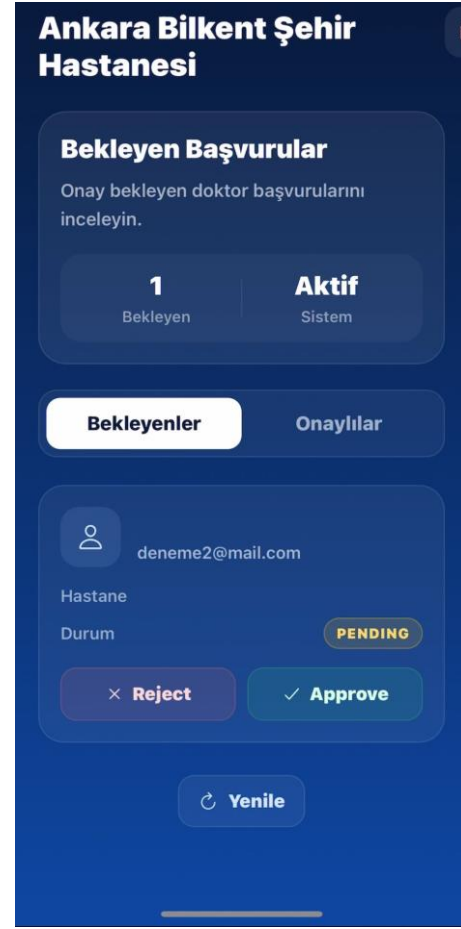


Figure 7: Pending Doctor Approval Dashboard

The second half of the access workflow is handled through the hospital panel. In Figure 6, the hospital administrator authenticates using the hospital identity and password stored in the backend database. Once access is granted, the hospital is redirected to a management dashboard where pending doctor requests are listed, as shown in Figure 7. This dashboard allows administrators to review applications and apply approval or rejection decisions directly from the mobile interface. The approval logic is tightly coupled with the backend verification process: once a physician is approved, the user record is updated so that the account becomes valid for future doctor login attempts. Through this interface, the frontend not only displays institutional records but also operationalizes the verification mechanism that governs access to the entire diagnostic platform.

4.4. End-to-End Clinical Workflow and Patient Management



Figure 8: Main Dashboard and Image Upload Interface

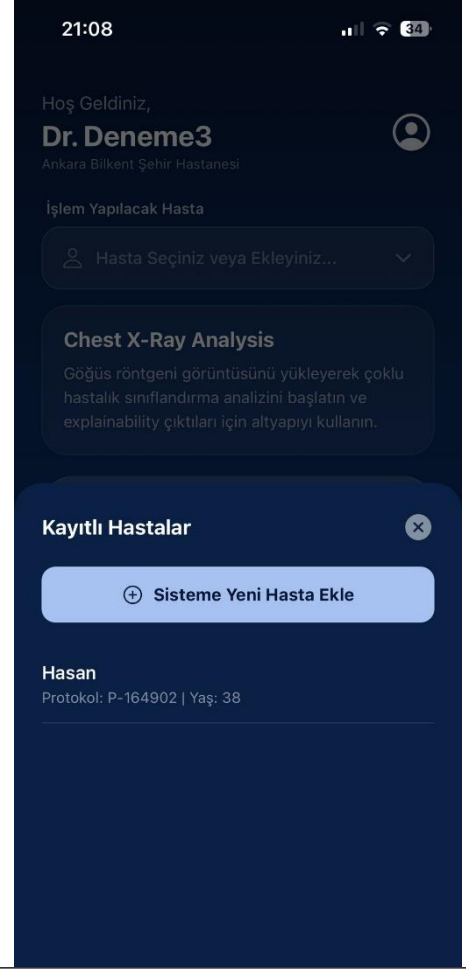


Figure 9: Patient Selection and Registration Management

The transition from a standalone deep learning model to a functional clinical tool was achieved through a structured mobile interface designed for real-world hospital environments. The system initiates with a secure patient selection and registration module, ensuring that every radiological analysis is tethered to a unique Protocol ID. As shown in the initial interface, physicians can either select from an existing patient database or register new entries with specific demographic details, providing a foundation for longitudinal medical tracking and data integrity.

4.5. Real-Time Neural Processing and Feedback Mechanisms



Figure 10: Real-time Analysis Progress

4.6. Explainable AI (XAI) and Diagnostic Visualization

A core achievement of this project is the implementation of Explainable AI (XAI) to bridge the gap between "black-box" neural networks and clinical trust. Beyond providing a list of potential pathologies, the system visualizes its findings by overlaying precise anatomical contours on the X-ray image. For instance, a high-probability detection like Effusion is not only reported as a percentage but is also localized using colored polygons, allowing the radiologist to immediately verify the model's focus area against anatomical landmarks.



Figure 11: Explainable AI Visualization with Disease Contours

4.7. Seamless Diagnostic Reporting and Patient Record Management



21:10

Consolidation %17.6
Eşik: 0.5

Fibrosis %16.6
Eşik: 0.5

Pleural_Thickening %14.1
Eşik: 0.55

Nodule %13.7
Eşik: 0.55

Pneumothorax %8.4
Eşik: 0.55

Pneumonia %8.3
Eşik: 0.35

Resmi Rapor Kaydı

Kurum: Ankara Bilkent Şehir Hastanesi

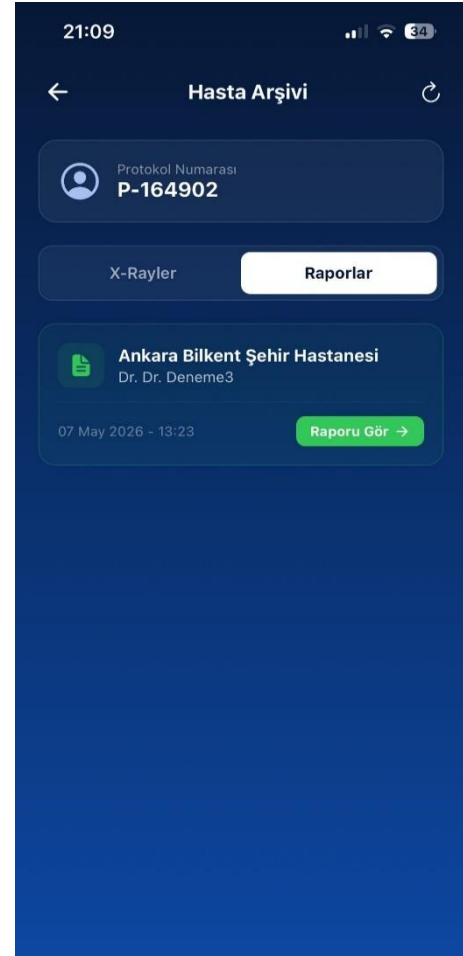
Hekim: Dr. Deneme3

Klinik değerlendirmenizi buraya yazın...

Raporu Onayla ve Arşivle ✓

Hasta Protokol: P-164902
Analiz Tarihi: 14 May 2026 - 21:10

Figure 12: Physician Reporting Interface and Clinical Note Entry



21:09

Hasta Arşivi

Protokol Numarası
P-164902

X-Rayler Raporlar

Ankara Bilkent Şehir Hastanesi
Dr. Dr. Deneme3

07 May 2026 - 13:23 **Raporu Gör** →

Figure 13: Patient Archive View for Finalized Medical Reports

The interface is designed to transform complex data into a streamlined reporting workflow, allowing the physician to finalize a diagnosis with minimal effort. As shown in the side-by-side comparison, once the analysis is complete, the doctor can easily review the results, add clinical notes, and generate a formalized report directly within the application. These reports are then instantly synchronized with the patient's dedicated profile. This integration ensures that the physician can access all historical diagnoses and previous X-ray reports through a simplified patient panel, facilitating a quick and organized review of the patient's medical journey without navigating through complex menus.

4.8. Integrated Digital Archiving and Diagnostic History



Figure 14: Longitudinal X-Ray Analysis History with Status Indicators



Figure 15: Main Dashboard Featuring Recent Diagnostic

The frontend architecture establishes a robust digital archiving system that centralizes all diagnostic activities for long-term patient management. As shown in the side-by-side interface comparison, the system automatically categorizes every finalized analysis into a structured "Patient Archive," where users can toggle between raw X-ray results and official medical reports. This dual-view organization is complemented by a "Recent Analyses" feed on the main dashboard, which provides immediate visibility into the most recent diagnostic outcomes, such as "Findings Detected" (marked in red) or "Clear/Normal" (marked in green). By unifying these two views, the interface allows physicians to monitor a patient's progress over time with high visual clarity, ensuring that critical historical data is always accessible for comparative evaluation.

4.9. Lessons Learned

The development of the mobile interface provided several insights into the importance of intuitive design in medical software:

- **The Power of Visual Transparency:** One of the most significant takeaways was that raw data is often overwhelming in clinical settings. By translating complex AI outputs into visual contours and color-coded markers, the interface reduces the doctor's cognitive load. Providing a clear visual "why" behind an AI suggestion is essential for building trust in digital diagnostic tools.
- **Context-Aware Interactivity:** Implementing conditional logic, where diagnostic selections only become active if the AI-detected probability exceeds specific thresholds, proved vital for preventing errors. This ensures that the physician only interacts with clinically significant findings, maintaining focus on the most relevant pathologies.
- **UX for High-Stress Environments:** The choice of a dark-themed UI and high-contrast status indicators (Red for findings, Green for normal results) demonstrated how design can directly support a radiologist's focus. A simplified, streamlined workflow minimizes the time spent on administrative tasks and maximizes time for clinical evaluation.
- **Structured Access and Verification:** Separating doctor and hospital access proved essential for improving role clarity and reducing confusion during login and onboarding. Implementing a hospital based approval mechanism increased system reliability by ensuring that newly registered doctors could not directly access clinical functions without institutional verification. Designing a simple and guided progression from registration to approval and login improved usability and helped users understand their current system status at each stage.

4.10. Future Work

The future roadmap for the application's frontend aims to enhance clinical efficiency and diagnostic precision through advanced interactive features. A primary objective is the transition from static image preprocessing to a Dynamic Augmentation Layer, allowing physicians to manually adjust parameters like contrast (CLAHE) and brightness (Gamma Correction) in real-time to better visualize subtle medical signs. Additionally, we plan to implement a Comparative Multi-View Interface supporting the simultaneous display of Posterior-Anterior (PA) and Anterior-Posterior (AP) views. To support the "physician-in-the-loop" model, future iterations will include an AI Feedback Loop, where doctor corrections and manual annotations are captured through the UI for continuous model refinement. Finally, we aim to integrate Predictive Trend Visualization, using historical archive data to generate visual charts of disease progression to help clinicians monitor treatment efficacy.

Additionally, future improvements for this module may include stronger account verification mechanisms such as e-mail confirmation, password recovery, and multi-factor authentication. The hospital approval panel can be extended with advanced filtering, search,

and status history features for better administrative control. Another possible enhancement is the integration of notification support, allowing doctors to be informed automatically when their accounts are approved or rejected. These additions will make the authentication workflow more secure, scalable, and user-friendly.

5. Database Design and Implementation

In this project, the database layer was one of the main components that supported the overall workflow of the Chest X-Ray Analysis System. The database was not only used for storing information, but also for connecting doctors, hospitals, patients, uploaded X-ray images, and analysis records in a structured way. MongoDB Atlas was selected as the cloud-based database solution because it provides a flexible document-based structure. This flexibility was useful for our project since medical records, doctor information, patient data, and analysis results may require new fields in future versions of the system.

The selection of MongoDB Atlas as the database system was based on several practical and technical factors. First, the team members responsible for the backend had prior experience with MongoDB, which reduced the learning curve and allowed faster development. Second, MongoDB Atlas provides a free-tier cloud hosting option, which made it accessible for an academic project without additional infrastructure cost. Third, MongoDB's document-based NoSQL structure was suitable for this project because different collections such as users, patients, and analyses have varying field requirements. A relational database with a fixed schema would have required additional migration steps each time a new field was needed, whereas MongoDB allowed these changes without restructuring the entire database. Finally, the availability of extensive documentation and community support made it easier to resolve issues during development.

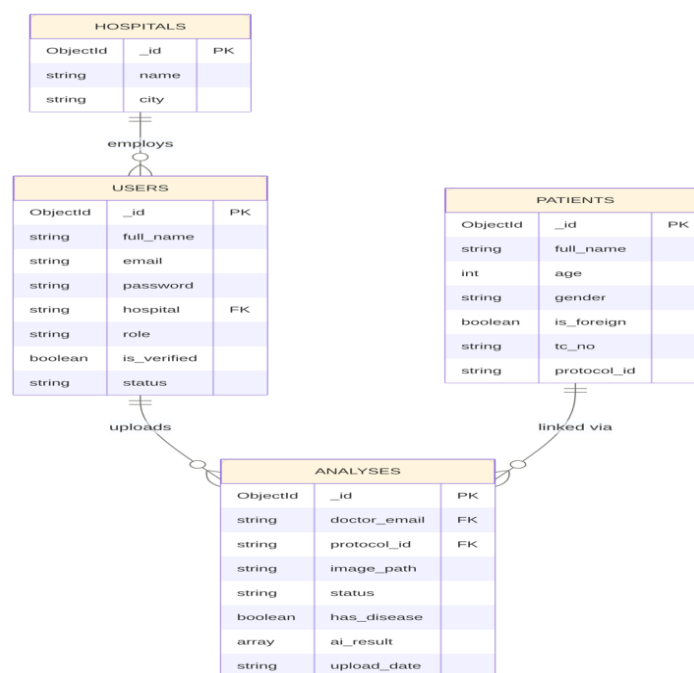


Figure 16: Collection schema of the MongoDB database.

The database was tested with approximately 25 to 50 records distributed across the four collections: users, patients, hospitals, and analyses. The “users” collection stores 8 fields per document including authentication and approval status fields, while the “patients” collection stores 7 fields covering demographic and identification information. The hospitals collection holds 4 fields per document, and the analyses collection stores up to 8 fields per record depending on the processing status. The total data size of the ceng408 database reached 24.75 MB during testing, with 4 indexes maintained across the collections. For completed analyses, the “ai_result” field stores an array of 14 disease classification scores, each representing a confidence value for a specific chest pathology such as Pneumonia, Cardiomegaly, or No Finding. This structure allows the system to preserve the full output of the AI model rather than only the top prediction, which supports more detailed clinical evaluation in future use.

Another contribution is the connection between patient records and analysis records through the protocol ID. When a patient is added to the system, a unique protocol ID is generated automatically. This protocol ID is later used while uploading an X-ray image, so the uploaded image can be linked to the correct patient. This design helps prevent confusion between patients and makes the analysis process more traceable. The patient module also checks whether a T.C. number already exists before creating a new patient, which helps reduce duplicate records.

The database also contributes to system security. During doctor registration, passwords are not stored directly as plain text. Instead, bcrypt is used to hash the password before saving it to the database. In addition, doctor accounts are first created with a pending status and are not allowed to log in until they are approved by an administrator. This approval mechanism is important because the system handles medical-related information, and unauthorized access should be prevented.

Another important contribution is the validation performed before saving an analysis record. Before an X-ray image is uploaded, the backend checks whether the entered protocol ID belongs to an existing patient. If there is no patient with that protocol ID, the system does not accept the image. The backend also checks whether the uploaded file is an image file. After these controls, the image path, doctor email, protocol ID, analysis status, AI result field, and upload date are saved in the analyses collection.

5.1. Applications

The database structure directly supports the real usage scenario of the Chest X-Ray Analysis System. A doctor can register to the system, wait for admin approval, log in after approval, add patients, upload chest X-ray images, and track analysis records. Each uploaded X-ray image can be connected to both the doctor who uploaded it and the patient it belongs to. This makes the system useful for medical environments where patient records and image analysis results must be stored in an organized way. Figure 1 illustrates this workflow and the corresponding database operations at each step.

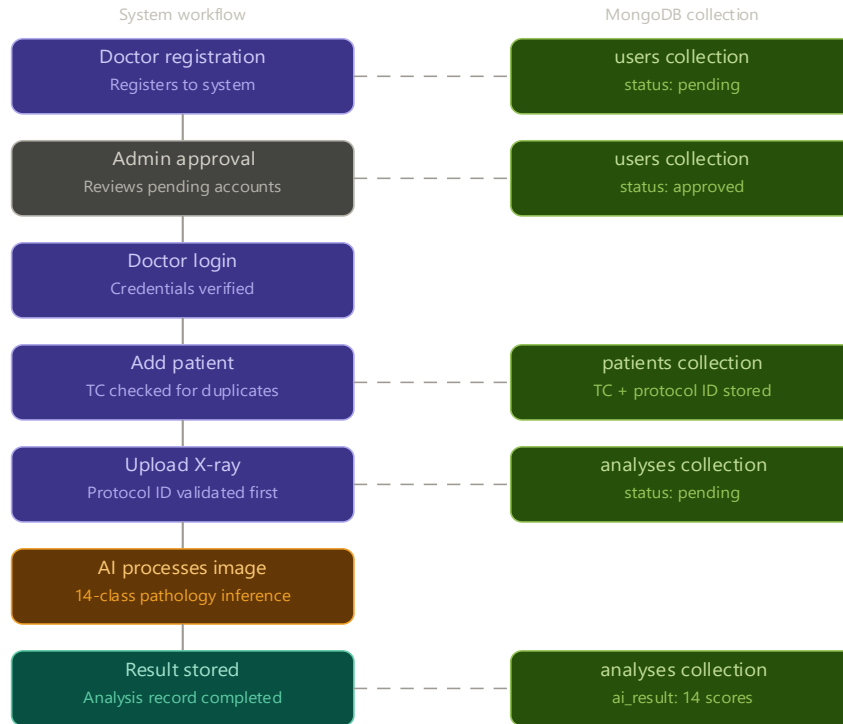


Figure 17: workflow and the corresponding database operations at each step

The hospital information stored in the database also supports the registration process. Since doctors are associated with hospitals, the system can later be extended for hospital-based filtering, hospital-specific doctor lists, or hospital-based analysis statistics. In this way, the current database design can support not only the existing application, but also more advanced healthcare management features in the future.

5.2. Lessons Learned

During the development of the database part, we learned that database design is not only about saving data. It also affects security, data consistency, and the usability of the whole application. One important lesson was the need to define clear relationships between different parts of the system. Since MongoDB is not a relational database like SQL databases, these relationships had to be managed carefully through fields such as email and protocol ID.

Another lesson was the importance of validation before inserting data into the database. For example, checking duplicate patients, checking whether a protocol ID exists before uploading an X-ray image, and checking the image file type helped make the system more reliable. Without these controls, the database could contain incorrect or unrelated records.

We also learned that authentication data must be handled carefully. Storing passwords directly would be unsafe, so password hashing became an important part of the database design. Similarly, the admin approval process showed us that user access should be controlled, especially in applications related to healthcare data.

One specific design decision that shaped the database structure was the use of two separate identification fields for patients. Initially, the T.C. identity number was considered sufficient for patient identification. However, since T.C. numbers are sensitive personal data and should not be exposed in general application flows such as image uploads or analysis tracking, a separate protocol ID field was introduced. The T.C. number is only used during patient registration to prevent duplicate records, while the protocol ID is used for all subsequent operations. This separation helped balance data privacy with system usability.

Another decision that was added during development was the doctor approval mechanism. As the system handles medical data, it became clear that allowing any registered user to immediately access patient records would be a security risk. Therefore, a status field was added to the users collection, and doctors are required to wait for administrator approval before they can log in. This change required updating both the registration and login logic, and showed that security requirements in healthcare applications often emerge more clearly once the full workflow is considered.

5.3. Future Work

As future work, the database layer can be improved in several ways. First, the MongoDB connection string should be moved from the source code to an environment file. This would make the system more secure because database credentials should not be written directly inside the code.

Second, indexes can be added to frequently searched fields such as email, protocol ID, and T.C. number. This would improve query performance, especially when the number of users, patients, and analysis records increases.

Third, the analysis collection can be expanded. Currently, the system stores the image path, analysis status, AI result field, and upload date. In the future, this structure can be improved by adding doctor comments, final diagnosis confirmation by the physician, and a structured report output. Although the AI model currently returns confidence scores for 14 pathology classes, the system does not yet support doctor annotations or official report generation linked to each analysis. This would allow the system to keep a more complete medical analysis history for each patient.

Finally, the database can be improved to support patient history tracking. For example, a doctor could view all previous X-ray analyses of the same patient through the protocol ID. This would make the system more useful in real clinical scenarios because doctors often need to compare current and previous medical records.

Overall, the database layer provides a secure, organized, and extendable foundation for the Chest X-Ray Analysis System. It supports user authentication, doctor approval, patient registration, hospital data management, X-ray upload control, and analysis tracking. Therefore, the database is not only a storage unit in this project; it is one of the main components that controls the flow and reliability of the whole system.